Year 7 Mathematics

Unit 2: Number Skills

2.3 Positive and Negative Numbers

Positive and Negative Numbers

STARTER · 5 MIN

Date: _____ **Title: Positive and Negative Numbers**

LEARNING OBJECTIVES

- To compare, order, add and subtract positive and negative numbers

STARTER *Answer in your book. No calculators.*

1. Which is colder: −3 °C or 2 °C?
2. Work out 7 − 10
3. What is 4 less than 1?
4. Put in order: 5, −1, 3, 0

Starter — Answers

STARTER

Tick or correct in green pen

1. **−3 °C**

2. **−3**

3. **−3**

4. **−1, 0, 3, 5**

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can recognise positive and negative numbers on a number line	☐
SILVER	I can compare and order a list of numbers that includes positives and negatives	☐
GOLD	I can explain my reasoning using a number line	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Positive and Negative Numbers

INTRODUCTION · 3 MIN

Integers are whole numbers: positive, negative or zero.

Number line

Numbers get bigger to the right.

−5 < −2: −5 is further left

Symbols

< means 'is less than', > 'is more than'.

−1 > −4

Adding

Add a positive number: move right.

−2 + 3 = 1

Subtracting

Subtract a positive number: move left.

3 − 7 = −4

Examples 1 and 2 — Comparing Negative Numbers

EXAMPLES · 5 MIN

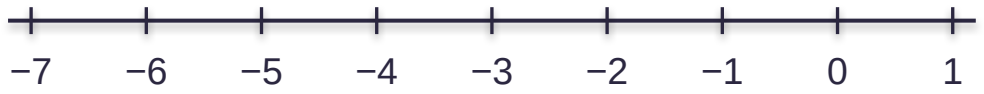
BRONZE

Watch and copy into your book

Hint Further right = bigger < less than > more than

Example 1

Write < or > between −6 and −2.



Example 2

Mark −4, 1 and −1 on a number line from −5 to 5.

Examples 1 and 2 — Corrections

CORRECTIONS

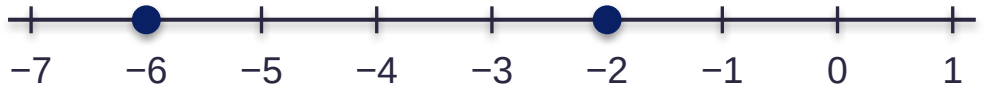
BRONZE

Check your copy and correct in green pen

Example 1

Further left is smaller

Answer = −6 < −2

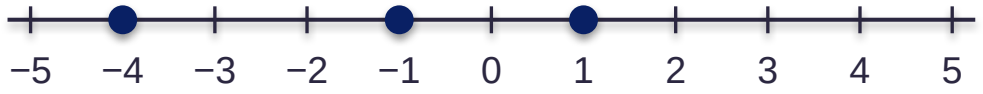


Tip: −6 °C is colder than −2 °C.

Example 2

Count from 0

Negative numbers go left of 0.



Tip: label 0 first.

Task 1

PRACTICE · 9 MIN

BRONZE *Write < or >, or answer the question*

Hint Further right = bigger < less than > more than

1. $-3 \square 2$

2. $-7 \square -4$

3. $0 \square -1$

4. $-10 \square -12$

5. Which is colder:
   -8 °C or -5 °C?

★ Which number is halfway between -6 and 2?

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1. $-3 < 2$
2. $-7 < -4$
3. $0 > -1$
4. $-10 > -12$
5. $-8\ °C$

★ -2

Examples 3 and 4 — Ordering, Adding and Subtracting

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Hint Add: move right Subtract: move left

Example 3

Write in ascending order:

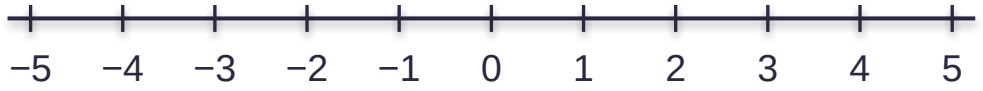
4, −6, 1, −3, 0

Example 4

Work out

(a) 3 − 7

(b) −2 + 3



Examples 3 and 4 — Corrections

CORRECTIONS

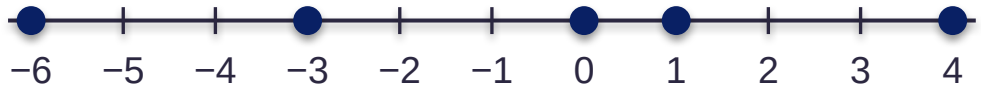
SILVER

Check your copy and correct in green pen

Example 3

Ascending: smallest first

−6, −3, 0, 1, 4



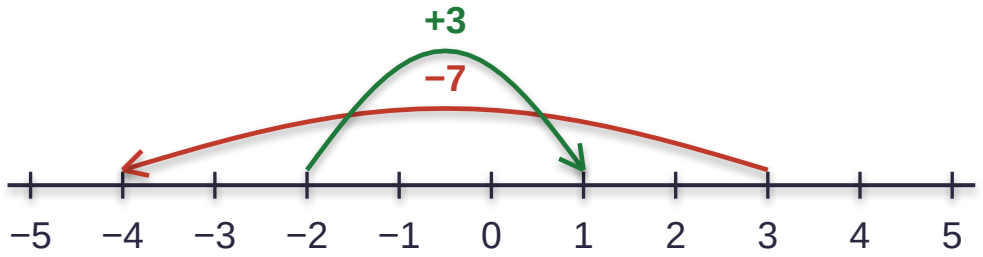
Tip: descending is biggest first.

Example 4

Start, then move left or right

(a) 3 − 7 = −4

(b) −2 + 3 = 1



Tip: subtract: left. Add: right.

Task 2

PRACTICE · 9 MIN

SILVER

Order the numbers or work out the answer

Hint Add: move right Subtract: move left

1 Ascending order:

2, −1, −4, 3

2 Descending order:

−5, 0, 4, −2

3 −6 + 10

4 2 − 9

5 −3 − 4

★ −5 + 8 − 6

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. **−4, −1, 2, 3**
2. **4, 0, −2, −5**
3. **4**
4. **−7**
5. **−7**

★ **−3**

Examples 5 and 6 — Negative Number Problems

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint Rise: add Fall: subtract Same signs +, different −

Example 5

(2 marks)

A freezer starts at −6 °C.

It rises by 11 °C, then falls by 9 °C.

Work out the final temperature.

Example 6

(2 marks)

Work out

(a) -3×4

(b) -2×-6

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

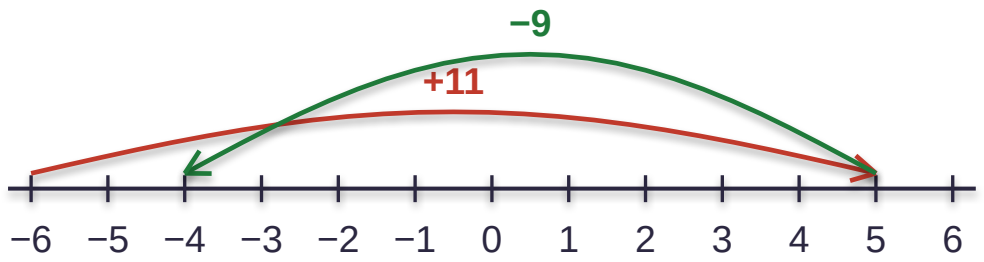
Example 5

Rise: add. Fall: subtract.

$-6 + 11 = 5$

$5 - 9 = -4$

Final = −4 °C



Tip: check on a number line.

Example 6

Signs: same +, different −

(a) $-3 \times 4 = -12$

(b) $-2 \times -6 = 12$

Tip: − × + = − and − × − = +

Task 3

PRACTICE · 9 MIN

GOLD

Exam-style: show all your working

Hint Rise: add Fall: subtract Same signs +, different −

1. It is −4 °C. It rises by 9 °C. New temperature?

2. It is 3 °C. It falls by 8 °C. New temperature?

3. −3 × 4

4. −5 × −2

5. Difference between −7 °C and 5 °C?

★ Find the missing number: □ × −6 = 42

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1. $-4 + 9 = 5\ °C$

2. $3 - 8 = -5\ °C$

3. -12

4. 10

5. $5 - (-7) = 12\ °C$

★ $\square = -7$

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Which is bigger: −9 or −4?

SILVER Work out −5 + 7

GOLD Work out −4 × −3

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE **−4**

SILVER **2**

GOLD **12**

Exam Practice — Comparing Numbers

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

Hint Further right = bigger
< less than > more than

Question 1

(2 marks)

Comparing temperatures.

(a) Which is warmer: −2 °C or −9 °C? (1)

.....

(b) Write < or >: −4 □ −1 (1)

.....

Question 2

(2 marks)

Work out each calculation.

(a) −6 + 10 (1)

.....

(b) 4 − 9 (1)

.....

Exam Practice — Ordering and Calculating

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint Add: move right Subtract: move left

Question 3

(2 marks)

Numbers: 4, −7, 0, −1, 2

(a) Write them in ascending order. (1)

.....

(b) Work out −3 − 5 (1)

.....

Question 4

(2 marks)

The temperature is −3 °C.

(a) It falls by 6 °C. New temperature? (1)

..... °C

(b) It rises by 10 °C from −3 °C. (1)

..... °C

Exam Practice — Negative Number Problems

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint Rise: add Fall: subtract
Same signs +, different −

Question 5

(2 marks)

A freezer starts at −9 °C.

Its temperature rises by 14 °C, then falls by 8 °C.

Work out the final temperature.

..... °C

Question 6

(2 marks)

Work out each calculation.

(a) −3 × 4 (1)

.....

(b) −6 × −5 (1)

.....

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

(a) −2 °C B1

(b) −4 < −1 B1

Tip: further right is bigger.

Question 2

(a) 4 B1

(b) −5 B1

Tip: use a number line.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

(a) $-7, -1, 0, 2, 4$ — B1

(b) -8 — B1

Tip: ascending means smallest first.

Question 4

(a) $-3 - 6 = -9\ °C$ — B1

(b) $-3 + 10 = 7\ °C$ — B1

Tip: falls: subtract. Rises: add.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

−9 + 14 = 5 — M1

5 − 8 = −3 °C — A1

Tip: do one change at a time.

Question 6

(a) −12 — B1

(b) 30 — B1

Tip: − × − = +